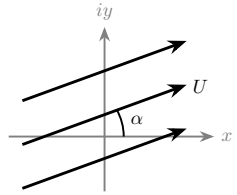
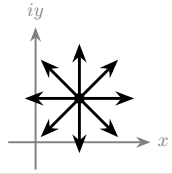
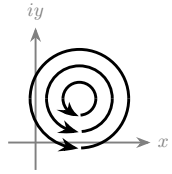
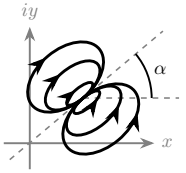
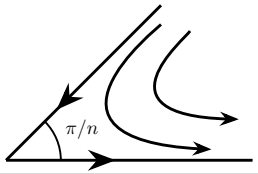
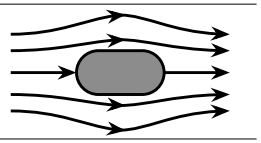
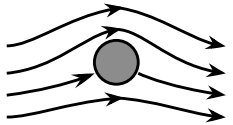
A Brief Handbook of Elementary Ideal Flows

Elementary flow	**Complex potential**, $F = \phi + i\psi$	**Potential**, $\phi(x,y)$ or $\phi(r,\theta)$	**Streamfunction**, $\psi(x,y)$ or $\psi(r,\theta)$	**Sketch of streamlines**
① Uniform flow at angle α to the horizontal	Az with $A = Ue^{-i\alpha}$ U is real but A is complex	$U(x\cos\alpha + y\sin\alpha)$ or $Ur\cos(\theta-\alpha)$	$U(y\cos\alpha - x\sin\alpha)$ or $Ur\sin(\theta-\alpha)$	
② Source ($Q>0$; for sink replace Q by $-Q$) located at $z_0 = x_0 + iy_0$	$\frac{Q}{2\pi}\log(z-z_0)$	$\frac{Q}{2\pi}\ln\sqrt{(x-x_0)^2+(y-y_0)^2}$ or, for $z_0=0$, $\frac{Q}{2\pi}\ln r$	$\frac{Q}{2\pi}\tan^{-1}\left(\frac{y-y_0}{x-x_0}\right)$ or, for $z_0=0$, $\frac{Q}{2\pi}\theta$	
③ Vortex (counterclockwise for $\Gamma>0$) located at $z_0 = x_0 + iy_0$	$\frac{-i\Gamma}{2\pi}\log(z-z_0)$	$\frac{\Gamma}{2\pi}\tan^{-1}\left(\frac{y-y_0}{x-x_0}\right)$ or, for $z_0=0$, $\frac{\Gamma}{2\pi}\theta$	$-\frac{\Gamma}{2\pi}\ln\sqrt{(x-x_0)^2+(y-y_0)^2}$ or, for $z_0=0$, $-\frac{\Gamma}{2\pi}\ln r$	
④ Doublet located at $z_0 = x_0 + iy_0$ and rotated by angle α	$\frac{me^{i\alpha}}{z-z_0}$ m is real	$\frac{m\cos\left(\tan^{-1}\left(\frac{y-y_0}{x-x_0}\right)-\alpha\right)}{\sqrt{(x-x_0)^2+(y-y_0)^2}}$ or, for $z_0=0$, $\frac{m\cos(\theta-\alpha)}{r}$	$-\frac{m\sin\left(\tan^{-1}\left(\frac{y-y_0}{x-x_0}\right)-\alpha\right)}{\sqrt{(x-x_0)^2+(y-y_0)^2}}$ or, for $z_0=0$, $-\frac{m\sin(\theta-\alpha)}{r}$	
⑤ Flow in a corner of angle π/n ($n>1/2$) $1<n<2$: flows past wedges of angle $2\pi\frac{n-1}{n}$ $n=2$: stagnation-point flow onto a horizontal plate	Az^n	[inconvenient to use Cartesian coords] or $Ur^n\cos(n\theta)$	[inconvenient to use Cartesian coords] or $Ur^n\sin(n\theta)$	
⑥ Flow past a Rankine body (a is a real number/horizontal shift) $①\{\alpha=0\}+②\{z_0=-a\}-②\{z_0=a\}$	$Uz + \frac{Q}{2\pi}\log\left(\frac{z+a}{z-a}\right)$	$Ux + \frac{Q}{2\pi}\ln\sqrt{\frac{(x+a)^2+y^2}{(x-a)^2+y^2}}$ [inconvenient to use polar coords]	$Uy + \frac{Q}{2\pi}\left[\tan^{-1}\left(\frac{y}{x+a}\right)-\tan^{-1}\left(\frac{y}{x-a}\right)\right]$ [inconvenient to use polar coords]	
⑦ Flow past a cylinder with circulation $①\{\alpha=0\}+④\{\alpha=0, m=UR^2, z_0=0\}-③\{z_0=0\}$	$U\left(z+\frac{R^2}{z}\right)+\frac{i\Gamma}{2\pi}\log z$	$U\left(1+\frac{R^2}{x^2+y^2}\right)x - \frac{\Gamma}{2\pi}\tan^{-1}\left(\frac{y}{x}\right)$ $U\left(1+\frac{R^2}{r^2}\right)r\cos\theta - \frac{\Gamma}{2\pi}\theta$	$U\left(1-\frac{R^2}{x^2+y^2}\right)y + \frac{\Gamma}{2\pi}\ln\sqrt{x^2+y^2}$ $U\left(1-\frac{R^2}{r^2}\right)r\sin\theta + \frac{\Gamma}{2\pi}\ln r$	

Inspired by Tables 12.1 and 12.2 of Granger's *Fluid Mechanics* (Dover Publications, 2012). Flows ① – ⑤ are the "building blocks" and the rest are superposition examples.